Software Engineering Lab-1

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Section - 5B

Lab 1: Requirements Table

Problem Statement #04: Student Project Portfolio & Showcase App, Campus & Academic Operations

Scenario: Annual project expos require a centralized showcase where student teams submit multimedia artifacts, judges grade projects using digital rubrics, and the public participates in voting with strict anti sybil protections. Stakeholders/Actors: Student Team, Panel Judge, Public Voter.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable student teams to create showcase profiles containing abstract summaries, demo video links, and verified GitHub repository URLs.	High	Pass: Profile is published to the showcase catalog upon validation of the repository link. Fail: An invalid URL format is accepted.	Establishes the core project record that judges grade and the public browses and votes on; without a verified repo link, project authenticity cannot be confirmed.
FR-002	Functional	The system shall allow a panel judge to score an assigned project against a weighted digital rubric (criteria: innovation, technical execution, presentation) and submit the score before the judging window closes.	High	Pass: Submitted rubric scores are recorded per criterion and locked from further edits once the judging window closes. Fail: A judge can submit an incomplete rubric or edit scores after the window closes.	Standardizes evaluation across judges and produces an auditable, tamper proof score that feeds the leaderboard.
FR-003	Functional	The system shall allow a verified public voter to cast exactly one vote per project during the live voting window.	High	Pass: A second vote attempt from the same verified identity for the same project is rejected with an on screen message. Fail: The system records more than one vote from the same identity for the same project.	Guarantees one person one vote fairness, which is essential to the legitimacy of the public vote component of the final ranking.
FR-004	Functional	The system shall verify a public voter's identity via a one time password (OTP) sent to a unique email address or phone number before accepting their first vote.	High	Pass: A vote is accepted only after the correct OTP is entered within a 5 minute validity window. Fail: A vote is accepted without OTP verification, or an expired OTP is accepted.	Core anti sybil control that prevents bots or duplicate accounts from being created solely to skew the public vote.
FR-005	Functional	The system shall display a live leaderboard combining normalized judge rubric scores	Medium	Pass: The leaderboard reflects a newly submitted vote or judge score, and correctly refilters by category, within the	Gives student teams, judges, and the public a transparent, trustworthy view of current

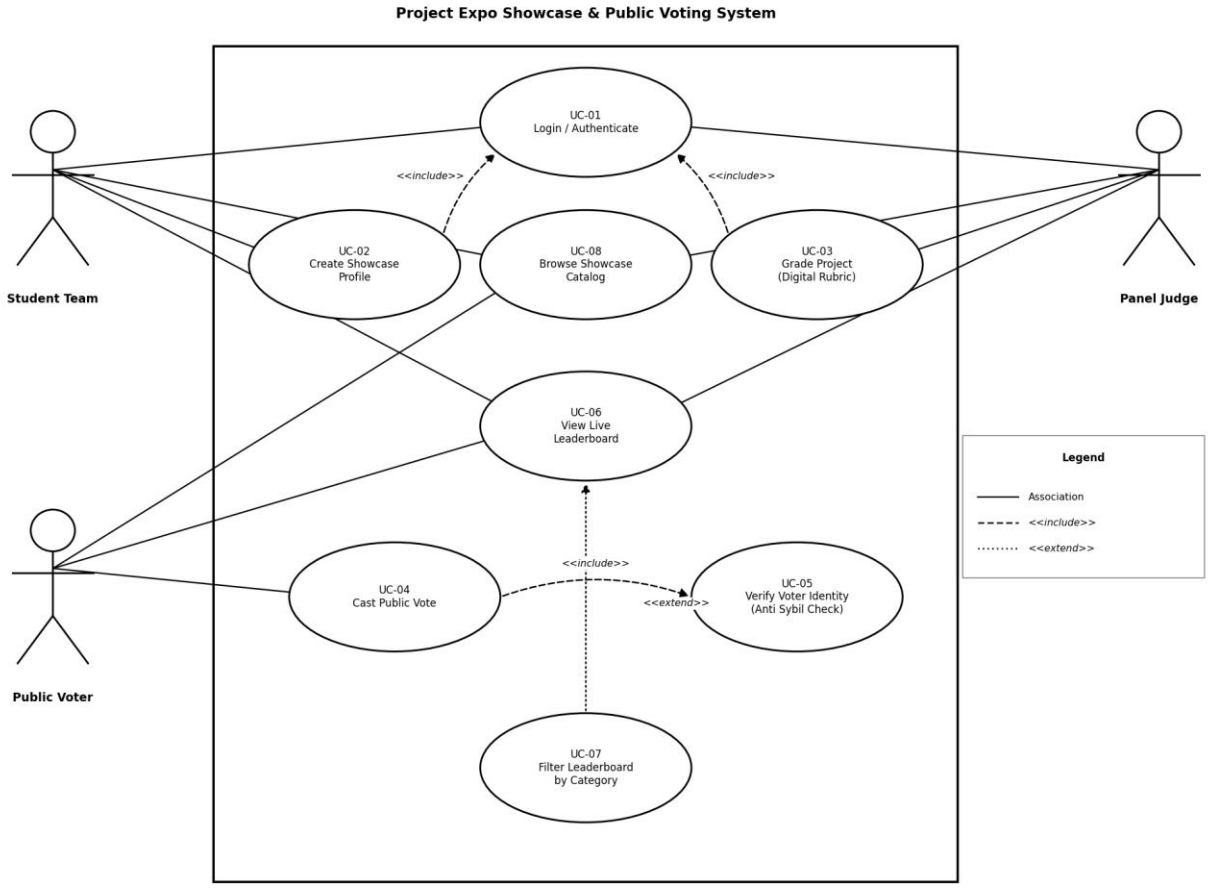
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		and public vote counts, filterable by expo category.		NFR-001 latency target. Fail: Stale standings are shown after new input has been recorded.	standings throughout the live event.
NFR-001	Nonfunctional (Performance & Security)	The public voting and leaderboard calculations shall update in real time with an API response latency under 300 ms.	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load.	Expo audiences expect near instant feedback on votes and rankings; noticeable lag undermines confidence in the results during a live event.
NFR-002	Nonfunctional (Security)	The system shall reject automated or duplicate voting attempts (bot traffic, repeated IP/device fingerprints, disposable email addresses) with at least 99.5% accuracy under simulated adversarial load, while blocking no more than 1% of legitimate voters.	High	Pass: A security test suite combining a simulated bot attack with a legitimate user sample confirms 99.5% or higher malicious vote rejection and 1% or lower false positive rate. Fail: Either threshold is missed.	Protects competition integrity from vote manipulation while ensuring genuine audience participation is not unfairly blocked.

Note: FR-001 and NFR-001 are the instructor provided baseline requirements (description and acceptance criteria reproduced as given); FR-002 to FR-005 and NFR-002 were elicited to complete the required set of 5 FRs and 2 NFRs for this scenario.

Lab 1: UML Use Case Diagram

Problem Statement #04: Student Project Portfolio & Showcase App

Actors: Student Team, Panel Judge, Public Voter. Includes at least one <<include>> and one <<extend>> relationship, as required.



Lab 1: Use Case Flow Specification

Problem Statement #04: Student Project Portfolio & Showcase App

Use Case: UC-04 Cast Public Vote

Primary Actor: Public Voter

Includes: UC-05 Verify Voter Identity (Anti Sybil Check)

Preconditions

The live voting window for the expo is open, the project showcase catalog is published, and the voter has a device with internet access. The voter has not already cast a vote for the selected project in this voting window.

Trigger

The public voter selects "Vote" on a project's showcase profile.

Main Success Scenario

1. Public voter browses the showcase catalog and opens a project's profile.
2. System displays the project profile with a "Vote" button, since this voter has not yet voted for this project.
3. Voter selects "Vote."
4. System checks whether the voter holds a verified session; if not, the Verify Voter Identity use case (UC-05) executes via «include»: the system sends a one time password (OTP) to the voter's email or phone.
5. Voter enters the OTP within the 5 minute validity window.
6. System validates the OTP and marks the voter's session as verified.
7. System records one vote for the selected project, tied to the voter's verified identity, and timestamps it.
8. System updates the live leaderboard and vote count for the project in real time (see NFR-001).
9. System displays a confirmation message: "Your vote has been recorded."
10. Use case ends successfully.

Alternate Flow

5a. Invalid or Expired OTP (Failed Identity Verification)

a. System displays "Incorrect or expired code, please try again."
b. Voter may request a new OTP, up to 3 resend attempts within the voting window.
c. If the voter enters a valid OTP within the allowed attempts, the flow resumes at main step 6.
d. If the voter exhausts all 3 attempts without a valid OTP, the system blocks further OTP requests from that email/phone for 15 minutes, displays "Too many attempts, please try again later," and the use case ends without a vote being recorded.

Postconditions

Success: Exactly one vote is stored for the (voter identity, project) pair; the project's vote count and the live leaderboard reflect the new vote.

Failure: No vote is recorded for the attempted (voter identity, project) pair; the voter's prior verification state, if any, is unaffected and they may retry within policy limits.